Student Name				
Teacher	GSL	MDI	HAL	SMH

CHEMISTRY

Unit 4, SAC 1: Organic Chemistry

Test on concepts from Chapters 10-14 of Heinemann Chemistry 2

Monday 17th of August, Period 4

Writing time: 70 minutes

Structure of Report

Section	Number of questions	Number of questions to be answered	Number of marks
A	10	10	10
B	3	3	32
			Total 42

- Students are permitted to bring into the assessment room: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
- Students are not permitted to bring into the examination room: blank sheets of paper and/or white out liquid/tape.

Materials supplied

- This question and answer book of **12 pages**.
- A data book

Instructions

- Write your name in the space provided above on this page.
- All responses must be written in English.
- Working out must be shown for all calculations
- Do NOT write on the data book

At the end of the assessment session

- Wait silently for the supervisor to collect your **question and answer booklet and data book.**

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

Students must not disclose the contents of the task; to do so will be a breach of VCE guidelines and will be dealt with according to VCAA regulations.

SECTION A: Multiple-choice questions

Instructions for Section A

Answer all questions in pencil on the answer sheet provided for multiple choice questions.
Choose the response that is correct or that best answers the question.
A correct answer scores 1, an incorrect answer scores 0.
Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Question 1

The number of structural isomers of C_4H_9Cl is:

- **A.** 2
- **B.** 3
- **C.** 4
- **D.** 5

Question 2

A major component of some nail polish removers is ethyl ethanoate. The correct semi-structural formula for ethyl ethanoate is:

- **A.** $CH_3COOCH_2CH_3$ (incorrectly marked in Ed Perfect)
- **B.** $CH_3CH_2COOCH_3$
- **C.** CH_3COOCH_3
- **D.** $CH_3CH_2COOCH_2CH_{31}$

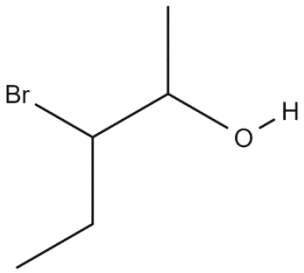
Question 3

The number of hydrogen atoms in a molecule of 2,4-dimethyl-4-hydroxypentanoic acid is:

- **A.** 12
- **B.** 13
- **C.** 14 (incorrectly marked in Ed Perfect)
- **D.** 15

Question 3

Consider the molecule



The IUPAC name for this molecule is:

- **A.** 2-bromo-1-methyl-butatan-1-ol
- **B.** 2-hydroxy-3-bromo-pentane
- **C.** 3-bromo-2-hydroxypentane
- **D.** 3-bromopentan-2-ol

Question 5

Hydrogen chloride gas can react with but-1-ene to produce 1-chlorobutane. This type of reaction is called:

A. substitution
B. oxidation
C. hydrogenation
D. addition

Question 6

The measurement that would be most useful in identifying an unknown substance separated by HPLC would be the:

A. total distance moved by the substance along the stationary phase
B. time taken for the substance to pass through the instrument
C. height of the peak produced by the substance
D. total area under the peak produced by that substance

Question 7

The low resolution H-NMR spectrum of the substance $CH_3CH(OH)CH_3$ would consist of:

A. three peaks
B. four peaks
C. two singlets and two triplets
D. one singlet, one doublet and one septet

Question 8

Two chemical containers have lost their labels. One contains ethanol and the other ethanoic acid. The most appropriate instrumental analysis to use to identify and determine the concentrations of the organic molecules is:

A. HPLC
B. ^{1}H-NMR spectroscopy
C. calorimetry
D. ^{13}C-NMR spectroscopy

Question 9

In a titration of a strong base with a strong acid, the following procedure was used:

1. A burette was filled with the standard acid.
2. A pipette was rinsed with some base solution.
3. A conical flask was rinsed with some base solution.
4. A pipette was used to transfer a measured volume of base solution into the conical flask.
5. Indicator was added to the base sample and it was titrated to the endpoint with the acid.

Which statement is correct?

A. The calculated base concentration will be correct
B. The calculated base concentration will be lower than the real value
C. The calculated base concentration will be higher than the real value
D. The endpoint will be the same as the equivalent point

Question 10

A redox titration is carried out by adding purple $KMnO_4$ oxidant solution from a burette to a solution of H_2O_2 in a conical flask, under acidic conditions. Which of the following would correctly describe the observed colour and the product formed in the conical flask before the equivalence point is reached?

	Observed Colour	Product Formed
A.	purple	H_2
B.	purple	O_2
C.	colourless	H_2
D.	colourless	O_2

END OF SECTION A

Section A: Multiple Choice Answer Sheet

This answer sheet must not be detached from this booklet.

INSTRUCTIONS FOR MULTIPLE CHOICE ANSWER SHEET

Use a **PENCIL** for **ALL** entries. For each question, shade the box which indicates your answer.
Marks will **NOT** be deducted for incorrect answers.
NO MARK will be given if more than **one** answer is completed for any question.
If you make a mistake, **ERASE** the incorrect answer - **DO NOT** cross it out.

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D

SECTION B Short Answer Questions

Instructions for Section B

Answer **all** questions in the spaces provided. Write using black or blue pen.
To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; un-simplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No marks will be given for an incorrect answer unless it is accompanied by details of the working.
- Make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_2(g)$; NaCl(s)

Question 1 (8 marks)

Butan-1-ol has the chemical formula C_4H_9OH.

a. Write the IUPAC name of three other possible isomeric alcohols with the formula C_4H_9OH. 3 marks

b. In the presence of an acidified solution of potassium dichromate ($K_2Cr_2O_7$), butan-1-ol reacts to produce an acidic product.

i. Write the semi-structural formula of the acidic product. 1 mark

ii. Write the complete balanced equation for the oxidation of butan-1-ol. 2 marks

Reduction: $6e^- + 14H^+(aq) + Cr_2O_7^{2-}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

Oxidation:

Overall:

iii. The presence of the acidic product can be confirmed by distilling and collecting the product, then testing it with solid sodium carbonate. Describe what would be expected to be observed when the acidic product is mixed with the solid sodium carbonate and write a chemical equation for the reaction. 2 marks

Observation:

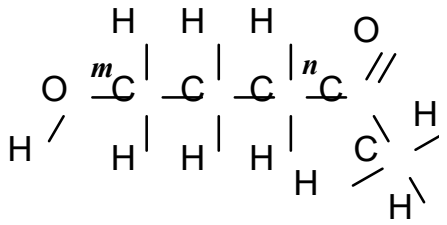
Chemical equation:

Question 2 (13 marks)

Compounds **A, B and C** are isomers of $C_5H_{10}O_2$.

a. Isomer **A** is a butyl ester. Draw the skeletal structure of **A**. 1 mark

b. Isomer **B** is the compounds shown below. Some of the carbons have been labelled (***m*** and ***n***).



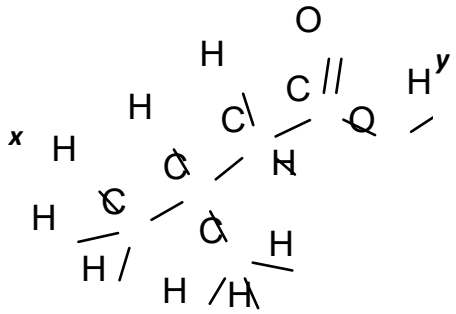
i. Deduce the number of peaks in the ^{13}C-NMR spectrum of Isomer **B**. 1 mark

ii. What is the expected chemical shift for the carbons labelled ***m*** and ***n***? 2 marks

Carbon ***m***:

Carbon ***n***:

c. Isomer **C** is the acid shown below. Some of the protons have been labelled.



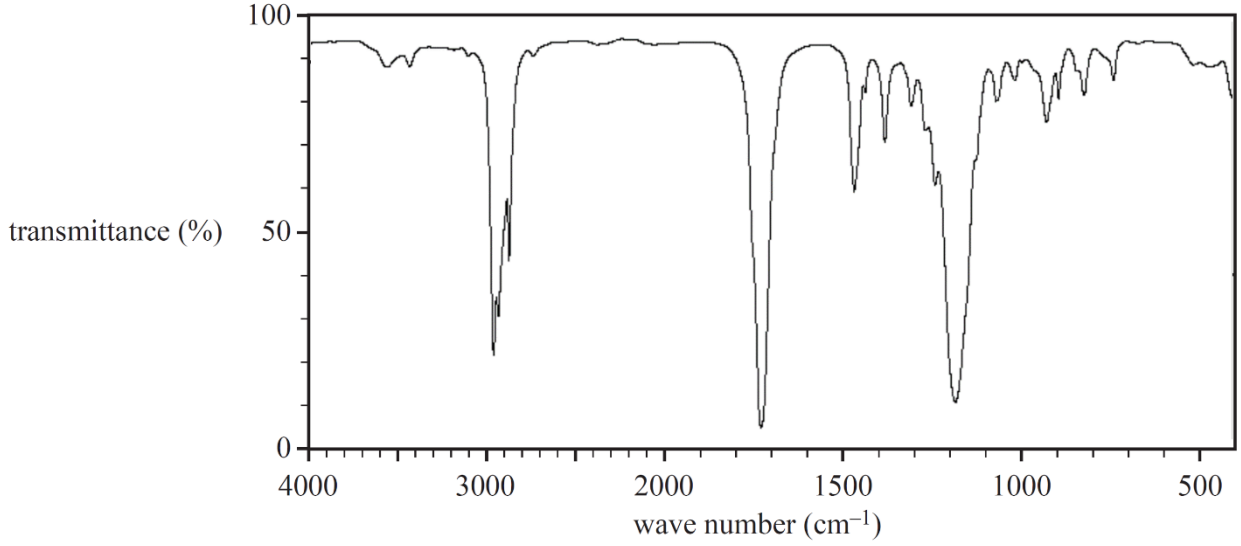
i. Deduce the number of peaks in the ^{1}H-NMR spectrum of Isomer **C**. 1 mark

ii. What is the expected splitting patter for the proton environments labelled ***x*** and ***y***? 2 marks

Proton environment ***x***:

Proton environment ***y***:

d. The IR spectrum of one of these isomers is shown below.



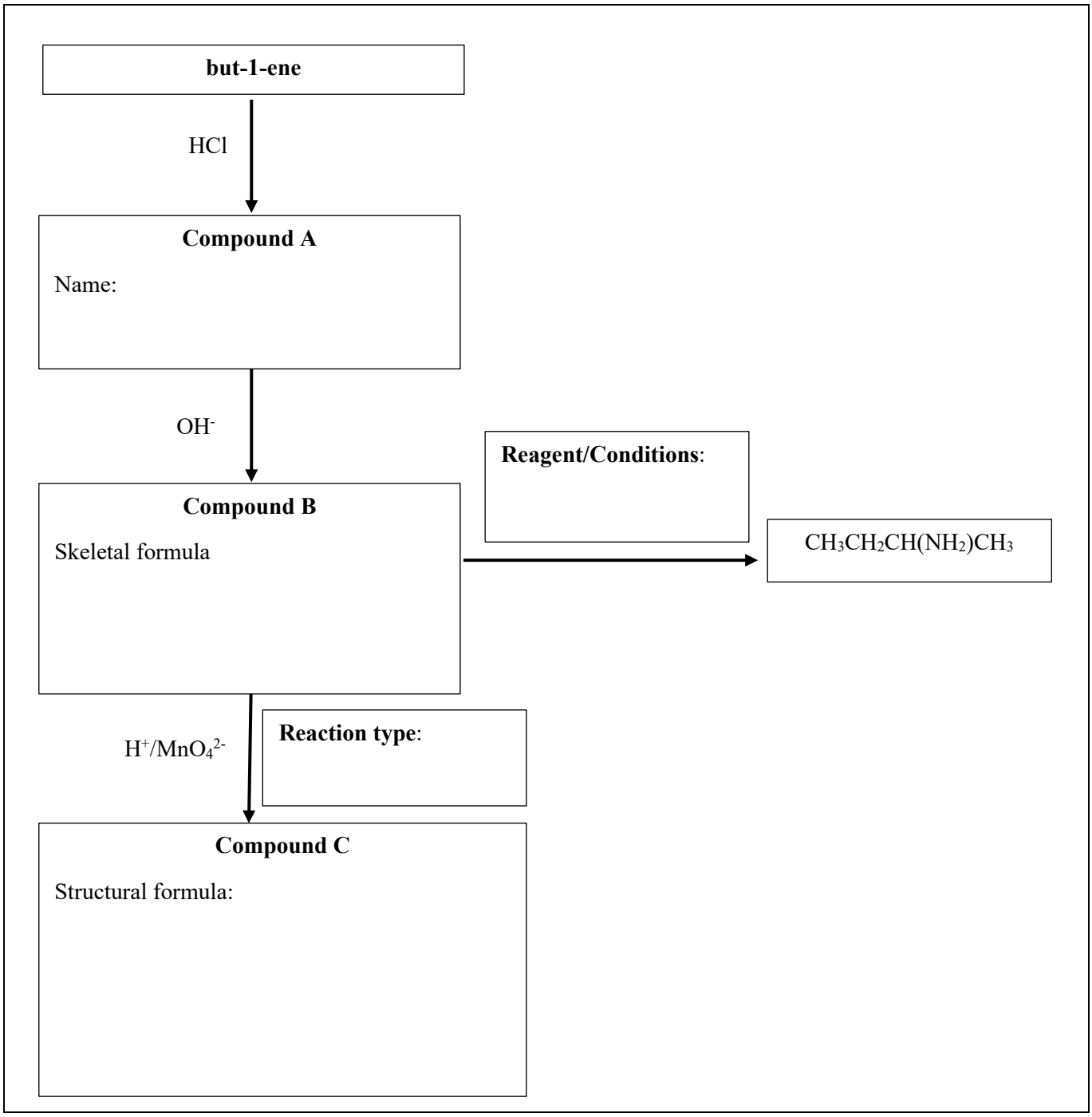
Data: SDBS Web, <http://sdbs.db.aist.go.jp>,
National Institute of Advanced Industrial Science and Technology

i. Describe the physical principles behind IR spectroscopy. 3 marks

ii. Identify which one of the three isomers, **A**, **B** or **C**, produced the IR spectrum above. Justify your answer with supporting evidence from the IR spectrum. 3 marks

Question 3 (11 marks)

a. The following diagram represents a reaction pathway for the synthesis of compound C from but-1-ene.



i. Give the IUPAC name of Compound A in the box provided. 1 mark

ii. When Compound A is reacted with hydroxide ions, Compound B is formed. Draw the skeletal formula for Compound B in the box provided. 1 mark

iii. Compound B can be converted into $CH_3CH_2CH(NH_2)CH_3$. Write the chemicals and conditions required for this conversion to occur in the box provided. 1 mark

iv. Identify the type of reaction occurring between Compound B and acidified permanganate (H+/MnO4-) in the box provided. 1 mark

v. Draw the structural formula of Compound C in the box provided. 1 mark

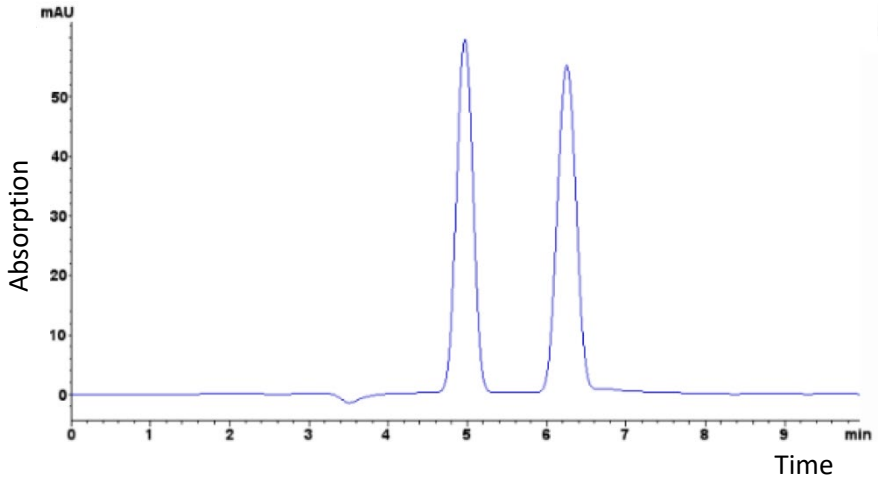
b. But-1-ene can be reacted with water, solid phosphoric acid and heat to produce two possible products, ***Y*** and ***Z***.

i. Give the IUPAC name for the both Compounds ***Y*** and ***Z***. 2 marks

Compound **Y**:

Compound **Z**:

The mixture of products (Compound ***Y*** and ***Z***) was separated using High Performance Liquid Chromatography (HPLC) with a polar stationary phase and a non-polar mobile phase. The following chromatogram with two peaks was obtained.



ii. Explain how the two compounds are separated as they travel through the HPLC column. 2 marks

iii. Explain how could you identify which peak corresponds to each compound. 2 marks

END OF TEST